NLP-BASED RESUME KEYWORD EXTRACTION AND JOB DESCRIPTION MATCHING USING TF-IDF

NOEL BIJU

III BCA B – 24UBC253

I. Problem Identification

The project aims to extract important keywords and skills from resumes and compare them with the requirements of a job description. This helps identify how well a candidate's resume matches a particular job using Natural Language Processing techniques.

Problem Relevance

This project is useful for simplifying the initial screening of resumes by identifying important skills and comparing them with job requirements. NLP is suitable because resumes and job descriptions are mainly unstructured text. The system can help recruiters identify candidates whose resumes are more relevant to a particular job.

II. Objectives

- To preprocess and analyse resume text using NLP techniques.
- To extract important keywords and skills from resumes.
- To represent resume text using TF-IDF.
- To compare resumes with a given job description using Cosine Similarity.
- To identify resumes that best match the job requirements.

III. Dataset

Dataset Name: AI Resume Analyzer – Job Role Prediction Dataset

Source: Kaggle

Dataset Size: 10,000 resume records

Attributes: Resume ID, Resume Text, Education, Experience Years, Skills, Job Role, and Category.

Target: Resume Text is used as the main text input for NLP processing.

Classes: The dataset contains different job roles and categories.

Limitations: The dataset contains synthetic resume data, so it may not fully represent the variety and complexity of real-world resumes.

IV. Methodology

The project follows these steps:

1. **Data Collection:** Resume data is loaded from the selected dataset.
2. **Data Preprocessing:** Resume text is converted to lowercase and unwanted characters and extra spaces are removed.
3. **TF-IDF:** TF-IDF is used to convert the resume text into numerical features and identify important terms.
4. **Keyword Extraction:** The highest TF-IDF scored terms are extracted as important keywords from each resume.
5. **Job Description Matching:** The given job description is converted into TF-IDF features and compared with the resumes.
6. **Similarity Calculation:** Cosine Similarity is used to calculate the matching score between the job description and each resume.
7. **Evaluation:** The resumes are ranked based on their matching scores to identify the best matches.

VI. Implementation

The project was implemented using Python in Google Colab. Pandas was used for loading and handling the dataset, while Scikit-learn was used for TF-IDF feature extraction and Cosine Similarity. Regular expressions were used for basic text preprocessing.

The resume text was cleaned and converted into TF-IDF features. Important keywords were extracted based on their TF-IDF scores. A sample job description was then processed using the same TF-IDF vectorizer, and Cosine Similarity was used to calculate the matching score for each resume. The resumes were finally ranked according to their similarity scores.

VII. Results

The system successfully extracted important keywords from the resumes and calculated their similarity with the given job description.

The average matching score was **4.30%**, while the highest matching score was **57.12%**. The best match was **Resume ID R007150**, with the job role **Data Scientist** and a matching score of **57.12%**.

The top five matching resumes achieved scores between **52.72% and 57.12%**, showing that the system can rank resumes based on their relevance to a given job description.

Qualitative Evaluation

The extracted keywords represent important terms and skills present in the resumes. Resumes containing more terms related to the given job description generally received higher similarity scores, allowing the system to rank resumes according to their relevance.

VIII. Limitations

- The dataset contains synthetic resume data and may not fully represent real-world resumes.
- The matching depends on the words present in the resume and job description.
- TF-IDF does not understand the deeper meaning or context of words.
- The system uses a sample job description, so results may vary for different job descriptions.
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IX. Future Scope

- Use larger and more diverse real-world resume datasets.
- Apply advanced NLP models such as BERT for better understanding of context.
- Develop a web-based interface for uploading resumes and job descriptions.
- Improve skill extraction and matching accuracy using semantic similarity.

X. Conclusion

The project successfully demonstrated the use of NLP for resume keyword extraction and job description matching. TF-IDF was used to identify important terms, while Cosine Similarity was used to measure the similarity between resumes and the job description. The system successfully ranked resumes based on their relevance, with the highest matching score of **57.12%**. The project shows that NLP can help simplify the process of identifying resumes relevant to specific job requirements.

GitHub Repository:

https://github.com/noelmulberry7-hash/NLP-Resume-Keyword-Extraction/tree/main